LESSON PLAN · SCRATCH JUNIOR (K–2) BEGINNER: LITTLE CODERS

Lesson 11: Short and Sweet Code

Module 4 | 30 minutes | Efficient code | KG–Grade 2 · ages 5–7

Objective: Use numbers and repeat blocks to solve puzzles with as few blocks as possible.

Success criteria

- I can count my blocks.
- I can use numbers and repeats to make code shorter.
- My code still works.

Key vocabulary

- **fewer**: Less of something.
- **pattern**: Something that repeats.
- **neat**: Short and tidy.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Shortcut game (5 min): turn “clap clap clap clap” into “clap 4”.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Shrink the code: learners follow the steps.
5 min	**Perform**	Snake path, Gem line, Shortest
3 min	**Probe**	Big square
4 min	**Wrap-up**	Exit ticket: Show me a short way to write ⬆️ ⬆️ ⬆️.

Unplugged warm-up

Shortcut game (5 min): turn “clap clap clap clap” into “clap 4”.

Teaching the idea

Efficiency: parameters and loops reduce block count. Encourage “make it work, then make it short”.

Analogy: Saying “three apples” instead of “apple, apple, apple”.

Live demo: Show a 10-block solution and shrink it to 3 blocks together.

Questions to ask

Ask	Listen for
How many blocks did you use?	Count.
Can a number help?	Yes.
Can a repeat help?	If there's a pattern.

Differentiation

Support
Solve first without a limit; then shrink together.

Stretch
Find a 2-block solution if one exists.

Common misconceptions

Misconception	How to fix it
Shortest code is always best for beginners	Working first, then short.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

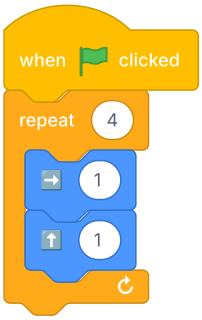
Exit ticket
Show me a short way to write ⬆️ ⬆️ ⬆️.

Homework
Find a pattern at home and say it the short way.

Answer key — Kitabash activities

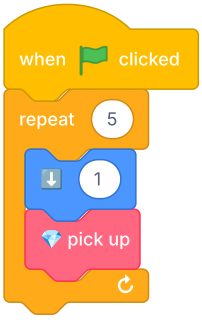
1. Snake path (perform · block code)

One solution:



2. Gem line (perform · block code)

One solution:



3. Shortest (perform · multiple choice)

Answer: ➡️ 5

4. Big square (probe · block code)

One solution:



Answer key — worksheet

1. **Key words:** fewer = Less of something. · pattern = Something that repeats. · neat = Short and tidy.
2. **Shortest:** A. ➡️ 5 — One block with 5.
3. **Make it short:** ➡️ 4; 🔁 3

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.